

A Phase-Triangular Case of the Block-GUE Fuglede–Kadison Determinant Conjecture

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Abstract

Mai and Speicher conjecture convergence of normalized determinants of arbitrary matrix-coefficient GUE pencils to the Fuglede–Kadison determinant of the free semicircular pencil. We prove the full conclusion for pencils which become upper triangular under fixed left/right equivalence and whose diagonal scalar pencils are real up to individual phases. This includes all real scalar pencils, all real simultaneously triangularizable pencils, and non-diagonal triangular systems. For arbitrary coefficients we also prove convergence of every fixed regularized determinant, isolating the remaining issue as uniform logarithmic integrability at the hard edge. The general conjecture remains open.

1 Source conjecture and normalization

Mai and Speicher’s Conjecture 1 on PDF page 17 of [4] is reproduced below.

FK-DETERMINANTS OF MATRIX-VALUED SEMICIRCULAR ELEMENTS 17

assume some special structure (in particular, quite some independence between the entries) for the matrices. Our setting is much more general and suggests thus the following conjecture for arbitrary Gaussian (or more general Wigner) random matrices.

Conjecture 1. Let $m \in \mathbb{N}$ and $a_i \in M_m(\mathbb{Z})$ be given and denote by η the corresponding completely positive map

$$\eta : M_m(\mathbb{C}) \rightarrow M_m(\mathbb{C}), \quad b \mapsto \eta(b) := \sum_{i=1}^n a_i b a_i^*.$$

Assume that η is rank non-decreasing. Then consider n independent GUE random matrices $X_1^{(N)}, \dots, X_n^{(N)}$ and let A_N denote the Gaussian block random matrix

$$A_N = \sum_{i=1}^n a_i \otimes X_i^{(N)}$$

of size $mN \times mN$. We claim that A_N is invertible for sufficiently large N and that

$$\lim_{N \rightarrow \infty} \Delta(A_N) = \text{cap}(\eta)^{\frac{1}{2m}} e^{-\frac{1}{2}},$$

both in expectation and almost surely.

Let $X_1^{(N)}, \dots, X_n^{(N)}$ be independent normalized GUE matrices, so that each empirical eigenvalue law tends to the standard semicircle law on $[-2, 2]$. For an $r \times r$ matrix T , write

$$\Delta_r(T) = |\det T|^{1/r}.$$

For $a_i \in M_m(\mathbb{C})$ put

$$A_N = \sum_{i=1}^n a_i \otimes X_i^{(N)}, \quad \eta_a(c) = \sum_{i=1}^n a_i c a_i^*,$$

and

$$\text{cap}(\eta_a) = \inf\{\det \eta_a(c) : c > 0, \det c = 1\}.$$

The source proves for $S_a = \sum_i a_i \otimes s_i$, with (s_i) a free standard semicircular family, that

$$\Delta(S_a) = \text{cap}(\eta_a)^{1/(2m)} e^{-1/2}. \quad (1)$$

Definition (phase-triangular). The tuple $(a_i)_{i=1}^n$ is *phase-triangular* if there are $P, Q \in GL_m(\mathbb{C})$ for which $b_i = P a_i Q$ is upper triangular for every i , and, for every $1 \leq j \leq m$, there are $\theta_j \in \mathbb{R}$ and $r_{1j}, \dots, r_{nj} \in \mathbb{R}$ such that

$$(b_i)_{jj} = e^{i\theta_j} r_{ij} \quad (1 \leq i \leq n). \quad (2)$$

Real simultaneously upper-triangular tuples are examples (take $P = I$ and Q to be the inverse change of basis). Allowing unrelated P, Q is a strict-equivalence version of the same flag structure.

2 Idea of the proof

Fixed left/right equivalence commutes with evaluation of the pencil and changes the normalized determinant by an explicit constant. After this equivalence, A_N is block upper triangular. Its determinant is therefore exactly the product of the determinants of its $N \times N$ diagonal blocks. Condition (2) makes each diagonal block a phase times a scalar multiple of a standard GUE matrix. The problem has thus become finitely many copies of the classical GUE log-determinant law; no independence between these copies is needed.

It remains to identify the resulting product with capacity. Apply the source formula (1) to the triangular free semicircular pencil. The Fuglede–Kadison determinant of an upper triangular operator matrix is the geometric mean of the determinants of its diagonal entries. Each diagonal entry is a scalar semicircular variable, whose determinant is $e^{-1/2}$ times its standard deviation. This gives the capacity product exactly.

3 The phase-triangular theorem

We use two standard determinant facts. First, if Z_N is normalized GUE, then the GUE log-determinant asymptotics (for example, the stronger fluctuation theorem of Bourgade–Mody [1]) imply

$$\Delta_N(Z_N) \longrightarrow e^{-1/2} \quad \text{almost surely and in } L^1. \quad (3)$$

Indeed, $N^{-1} \mathbb{E} \log |\det Z_N| \rightarrow -1/2$ and $\text{Var}(\log |\det Z_N|) = O(\log N)$; Chebyshev and Borel–Cantelli yield the almost-sure assertion. Uniform integrability of $\Delta_N(Z_N)$ follows from

$$\Delta_N(Z_N)^2 \leq \frac{1}{N} \text{Tr}(Z_N Z_N^*)$$

and the uniform Gaussian moment bounds.

Second, for an upper triangular $T = (t_{jk}) \in M_m(\mathcal{M})$ whose diagonal entries are injective and of determinant class,

$$\Delta_{M_m(\mathcal{M})}(T) = \left(\prod_{j=1}^m \Delta_{\mathcal{M}}(t_{jj}) \right)^{1/m}. \quad (4)$$

This is the triangular Fuglede–Kadison determinant identity; see the determinant-class discussion in [3]. One may also see it directly in the algebra of affiliated operators: factor out the diagonal matrix, leaving a unit upper-triangular matrix. The latter is a product of elementary unipotents and has determinant one; extended determinant multiplicativity gives (4).

Theorem 1. *Let $a_1, \dots, a_n \in M_m(\mathbb{C})$ be phase-triangular, and suppose that η_a is rank non-decreasing. Then A_N is invertible almost surely for every N , and*

$$\Delta_{mN}(A_N) \longrightarrow \text{cap}(\eta_a)^{1/(2m)} e^{-1/2} \quad (5)$$

almost surely and in L^1 . Consequently the convergence also holds in expectation. In particular this proves the full conclusion of Conjecture 1 for every phase-triangular integer tuple.

Proof. Choose P, Q and $b_i = Pa_iQ$ as in the definition, and set

$$B_N = \sum_i b_i \otimes X_i^{(N)} = (P \otimes I_N) A_N (Q \otimes I_N).$$

The matrix B_N is block upper triangular. Its j th diagonal block is

$$Y_{j,N} = \sum_i (b_i)_{jj} X_i^{(N)} = e^{i\theta_j} \sqrt{v_j} Z_{j,N}, \quad v_j = \sum_i r_{ij}^2,$$

where $Z_{j,N} = \sum_i r_{ij} X_i^{(N)} / \sqrt{v_j}$ is marginally standard GUE. Every v_j is positive: otherwise the triangular free pencil $\sum_i b_i \otimes x_i$ has a zero diagonal entry and cannot have full free rank, contrary to the rank-non-decrease criterion recalled in [4]. Thus each $Y_{j,N}$, and hence B_N and A_N , is almost surely invertible.

Finite block-triangular factorization gives

$$\Delta_{mN}(B_N) = \left(\prod_{j=1}^m \Delta_N(Y_{j,N}) \right)^{1/m}. \quad (6)$$

By (3), simultaneously for the finitely many j ,

$$\Delta_{mN}(B_N) \longrightarrow e^{-1/2} \left(\prod_{j=1}^m v_j \right)^{1/(2m)}$$

almost surely. The same convergence holds in L^1 : the geometric mean in (6) is at most the arithmetic mean, and the proof of (3) gives uniform L^2 bounds for its finitely many factors. Moreover,

$$\Delta_{mN}(B_N) = |\det P \det Q|^{1/m} \Delta_{mN}(A_N). \quad (7)$$

We now identify the constant. Let $S_b = \sum_i b_i \otimes s_i$. It is upper triangular with diagonal entries $e^{i\theta_j} \sqrt{v_j} s'_j$, where each s'_j is a standard semicircular variable. Since $\Delta(s'_j) = \exp \int \log |t| d\mu_{\text{sc}}(t) = e^{-1/2}$, (4) and the source formula give

$$\text{cap}(\eta_b)^{1/(2m)} e^{-1/2} = \Delta(S_b) = e^{-1/2} \left(\prod_j v_j \right)^{1/(2m)}.$$

Hence $\text{cap}(\eta_b) = \prod_j v_j$.

Finally, directly from the capacity definition,

$$\text{cap}(\eta_b) = |\det P \det Q|^2 \text{cap}(\eta_a). \quad (8)$$

Indeed $\eta_b(c) = P\eta_a(QcQ^*)P^*$, and $|\det Q|^{-2/m} QcQ^*$ runs through all positive determinant-one matrices as c does. Combining (7)–(8) proves (5). $\square$

4 Two general reductions

The first reduction gives genuine invertibility progress without a triangular hypothesis. Let $\mathcal{A} = \text{span}\{a_1, \dots, a_n\}$.

Proposition 2 (blow-up subsequence). *Suppose η_a is rank non-decreasing. There is a positive integer d (one may take $d \leq m - 1$ when $m > 1$ by [2]) such that, for every $k \geq 1$, A_{kd} is invertible almost surely.*

Proof. Rank non-decrease is equivalent to full noncommutative rank. Thus some d th blow-up of $\mathcal{A}$ contains an invertible matrix, and direct sums give an invertible evaluation in every blow-up of size kd . Consequently the polynomial $\det(\sum_i a_i \otimes Z_i)$ on $M_{kd}(\mathbb{C})^n$ is nonzero. Hermitian tuples form a real form, hence are Zariski dense in the complex tuple space, so the restriction of this polynomial to Hermitian tuples is still nonzero. Its zero set has Lebesgue measure zero there, while a GUE tuple has a smooth positive density. Therefore A_{kd} is almost surely invertible. $\square$

The second reduction holds in every size and identifies the analytic gap. For $\varepsilon > 0$ set

$$\Delta_{N,\varepsilon}(A_N) = \det(A_N A_N^* + \varepsilon^2 I_{mN})^{1/(2mN)}.$$

Proposition 3 (Regularized general reduction). *For arbitrary fixed $a_i \in M_m(\mathbb{C})$ and every $\varepsilon > 0$,*

$$\Delta_{N,\varepsilon}(A_N) \longrightarrow \exp\left\{\frac{1}{2}(\text{tr}_m \otimes \tau) \log(S_a S_a^* + \varepsilon^2 I)\right\} \quad (9)$$

almost surely and in L^1 . If η_a is rank non-decreasing, the right-hand side decreases, as $\varepsilon \downarrow 0$, to $\text{cap}(\eta_a)^{1/(2m)} e^{-1/2}$.

Proof. Almost-sure asymptotic freeness of independent GUE matrices gives convergence of the empirical spectral law of $A_N A_N^*$ to that of $S_a S_a^*$, together with an almost-sure common compact upper bound on the spectra. For fixed ε , the function $t \mapsto \frac{1}{2} \log(t + \varepsilon^2)$ is continuous on this compact interval, proving (9). The arithmetic–geometric mean bound and uniform Gaussian operator/Hilbert–Schmidt moment bounds give uniform integrability, hence L^1 convergence. Monotone convergence in the logarithmic functional calculus gives the $\varepsilon \downarrow 0$ limit, and (1) identifies it. $\square$

5 Hard-edge obstruction and scope

If $s_1(A_N), \dots, s_{mN}(A_N)$ are the singular values, the difference between regularized and actual logarithmic determinants is exactly

$$\log \Delta_{N,\varepsilon}(A_N) - \log \Delta_{mN}(A_N) = \frac{1}{2mN} \sum_{k=1}^{mN} \log\left(1 + \frac{\varepsilon^2}{s_k(A_N)^2}\right).$$

Thus Proposition 3 reduces the full conjecture to showing that this nonnegative quantity vanishes uniformly in the large- N limit as $\varepsilon \downarrow 0$, almost surely and in mean. This is precisely a hard-edge uniform-integrability estimate. Strong convergence controls the top edge, while generic polynomial anti-concentration loses a factor growing with the determinant degree mN . Existing matrix-Dyson local laws for Kronecker ensembles require stability or flatness hypotheses not verified here for every merely rank non-decreasing covariance map.

Accordingly, Theorem 1 is a solved structural subcase and Propositions 2–3 are general reductions; they do not settle arbitrary coefficient tuples. Bounded run-index and web searches found no exact later solution or duplicate of the phase-triangular statement. The novelty claim remains provisional pending specialist review.

References

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